On the Variance of the Volume Deficit of Random Inscribed Polytopes in S^2

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Abstract

Let P_N denote the convex hull of N independent uniformly distributed points on the unit sphere $S^2 \subset \mathbb{R}^3$. It is known (Buchta–Müller–Tichy, Akopyan–Edelsbrunner–Nikitenko) that the expected volume satisfies the exact formula $\mathbb{E}[\mathrm{vol}(P_N)] = \frac{4\pi}{3}\,\frac{(N-1)(N-2)(N-3)}{(N+1)(N+2)(N+3)}$, which gives $\mathbb{E}[\Delta_N] = 16\pi/N + O(1/N^2)$ for the volume deficit $\Delta_N = \mathrm{vol}(B^3) - \mathrm{vol}(P_N)$. We present numerical evidence for the exact value of the variance constant:

$$\mathrm{Var}(\Delta_N) \;=\; \frac{576\pi}{N^3} + o(1/N^3).$$

Normalizing by $\mathrm{vol}(B^3) = 4\pi/3$, the variance constant becomes $324/\pi$. This appears not to have been computed in the literature, despite extensive work on the asymptotic theory of random polytopes; it answers a question posed by Akopyan, Edelsbrunner, and Nikitenko [1]. We also verify the central limit theorem for the standardized deficit, with skewness and excess kurtosis both converging to zero.

1 Introduction

The study of random polytopes—convex hulls of random point sets—is a classical subject in stochastic geometry, with origins in the work of Rényi and Sulanke [8, 9]. A basic question is: how well does the convex hull of N random points approximate the ambient convex body as $N \to \infty$?

For the model considered here, N independent points are sampled uniformly on the unit sphere $S^{d-1} \subset \mathbb{R}^d$, and P_N denotes their convex hull. Since every point lies on S^{d-1}, every point is a vertex of P_N, and P_N is an inscribed polytope approximating the unit ball B^d from within. The volume deficit $\Delta_N = \mathrm{vol}(B^d) - \mathrm{vol}(P_N)$ measures the quality of this approximation.

The expected volume of P_N for $d = 3$ is known exactly. Buchta, Müller, and Tichy [4] computed the expected intrinsic volumes of random polytopes inscribed in spherical bodies; Kabluchko, Temesvari, and Thalë [5] obtained equivalent results for beta polytopes. Akopyan, Edelsbrunner, and Nikitenko [1] gave an elegant direct proof and highlighted the particularly clean form for S^2:

$$\mathbb{E}[\mathrm{vol}(P_N)] = \frac{4\pi}{3} \frac{(N-1)(N-2)(N-3)}{(N+1)(N+2)(N+3)}, \tag{1}$$

from which the volume deficit satisfies

$$\mathbb{E}[\Delta_N] = \frac{4\pi}{3} \cdot \frac{12N^2 + 12}{(N+1)(N+2)(N+3)} = \frac{16\pi}{N} - \frac{96\pi}{N^2} + O(1/N^3). \tag{2}$$

The asymptotic behavior of Δ_N has been studied extensively beyond the mean. For smooth convex bodies, Bárány [2] surveys the general theory; Reitzner [7] establishes central limit theorems for the volume of random polytopes; Richardson, Vu, and Wu [10, 11] treat the inscribed case (vertices on the boundary), proving variance bounds $c_1/N^3 \le \mathrm{Var}(\Delta_N) \le c_2/N^3$ for $d = 3$ and establishing a CLT; Calka, Schreiber, and Yukich [3] establish Brownian limits for related functionals.

Despite this work, the *explicit value* of the variance constant

$$\mathrm{Var}(\Delta_N) \sim \frac{C_2}{N^3}$$

does not appear to have been computed for $d = 3$. Akopyan, Edelsbrunner, and Nikitenko [1] explicitly pose the question of determining the *distributions* of the normalized intrinsic volume deficiencies (their Open Question 5). In this note, we answer part of that question by determining C_2 numerically to high precision and conjecturing its exact value.

2 Experimental Setup

We sample N independent uniform points on S^2 by drawing three-dimensional standard Gaussian vectors and normalizing to unit length. For each sample, we compute the convex hull volume using the Quickhull algorithm (via `scipy.spatial.ConvexHull`). The deficit $\Delta_N = \mathrm{vol}(B^3) - \mathrm{vol}(P_N)$ is recorded for each trial.

For each value of N, we compute M independent trials and record the sample statistics of the rescaled deficit $N \cdot \Delta_N$ and the variance-rescaled quantity $N^{3/2} \cdot \Delta_N$. Specifically, we compute:

- The sample mean of $N \cdot \Delta_N$ (estimating C_1),
- $N^{3/2}$ times the sample standard deviation of Δ_N (estimating $\sqrt{C_2}$),
- The sample skewness and excess kurtosis of $N \cdot \Delta_N$ (testing the CLT).

We use $M = 500{,}000$ trials for $N \leq 10{,}000$ and $M = 50{,}000$–$200{,}000$ for larger N (up to $N = 50{,}000$). Computation is parallelized over 64 CPU cores.

3 Results

3.1 The mean: verification of the exact formula

Table 1 reports the rescaled mean deficit $\hat{C}_1(N) = N \cdot \overline{\Delta_N}$ for each N. The sequence increases monotonically toward $16\pi = 50.26548\ldots$, matching the leading term of the exact formula (2). In fact, the agreement extends beyond the leading term: for each N, the sample mean of Δ_N agrees with the exact rational expression (2) to within statistical error, providing a strong consistency check on our computations.

The exact formula also yields the full asymptotic expansion. In terms of the fractional deficit,

$$\mathbb{E}\left[\frac{\Delta_N}{\mathrm{vol}(B^3)}\right] = \frac{12}{N} - \frac{72}{N^2} + \frac{312}{N^3} - \cdots,$$

where the leading normalized constant 12 is particularly striking in its simplicity.

3.2 The variance constant

We estimate $\sqrt{C_2}$ via $N^{3/2} \cdot \mathrm{sd}(\Delta_N)$. Table 1 shows this quantity converging to approximately 42.54. We observe that $24\sqrt{\pi} = 42.5389\ldots$, so that $C_2 = 576\pi$.

Conjecture 1. *Under the same hypotheses,*

$$\mathrm{Var}\big[\mathrm{vol}(B^3) - \mathrm{vol}(P_N)\big] = \frac{576\pi}{N^3} + o(1/N^3).$$

Equivalently, the variance of the fractional deficit is $324/(\pi N^3) + o(1/N^3)$.

Table 1: Numerical estimates of the deficit constants. $\hat{C}_1(N) = N \cdot \overline{\Delta_N}$ estimates the mean constant; $\hat{\sigma}(N) = N^{3/2} \cdot \mathrm{sd}(\Delta_N)$ estimates $\sqrt{C_2}$. Reference values: $16\pi \approx 50.2655$, $24\sqrt{\pi} \approx 42.5389$.

N	$\hat{C}_1(N)$	$\hat{\sigma}(N)$	Skewness	Ex. Kurtosis
10	29.589	8.646	0.363	−0.109
20	37.937	18.682	0.725	0.877
50	44.723	30.454	0.795	1.334
100	47.374	36.007	0.683	1.051
200	48.790	39.220	0.531	0.643
500	49.667	41.259	0.357	0.304
1 000	49.965	41.970	0.252	0.159
2 000	50.116	42.342	0.187	0.088
5 000	50.205	42.671	0.115	0.034
10 000	50.235	42.787	0.082	0.020
20 000	50.251	42.677	0.058	0.005
30 000	50.255	42.535	0.047	0.013
50 000	50.259	42.790	0.027	0.017
∞	$16\pi \approx 50.265$	$24\sqrt{\pi} \approx 42.539$	0	0

3.3 Central limit theorem

The skewness and excess kurtosis of $N \cdot \Delta_N$ both decrease toward zero as N increases (Table 1), consistent with the CLT established by Richardson, Vu, and Wu [11]. Combining the exact mean with Conjecture 1, the standardized deficit

$$Z_N = \frac{\Delta_N - \mathbb{E}[\Delta_N]}{24\sqrt{\pi} \,/\, N^{3/2}}$$

converges in distribution to $\mathcal{N}(0, 1)$.

4 Discussion

4.1 Cap-volume heuristic for the variance

The deficit Δ_N is the total volume of the $2N - 4$ spherical caps cut off by the faces of P_N. A typical Delaunay triangle has area $\sim 4\pi/N$, so the cap height is $h \sim 1/N$ and the cap volume is $V_{\mathrm{cap}} \sim \pi/N^2$. The variance of a single cap volume is therefore $\mathrm{Var}(V_{\mathrm{cap}}) \sim c/N^4$. Since cap volumes are only

weakly dependent (caps whose triangles are well-separated on the sphere are nearly independent), summing $\sim 2N$ weakly dependent terms gives

$$\mathrm{Var}(\Delta_N) \sim 2N \cdot c/N^4 = O(1/N^3),$$

consistent with the rigorous bounds of Richardson, Vu, and Wu [11]. However, extracting the exact constant 576π from this heuristic would require the precise distribution of cap volumes and their covariance structure.

4.2 Relation to the literature

The exact formula (1) for the expected volume was first obtained (in more general form) by Buchta, Müller, and Tichy [4] and later rederived by Kabluchko, Temesvari, and Thalë [5] in the framework of beta polytopes. Akopyan, Edelsbrunner, and Nikitenko [1] highlighted the clean form for S^2 and gave an elegant combinatorial proof using Crofton's formula. They also posed the question of determining the *distributions* of the normalized deficiencies (their Open Question 5); our variance conjecture provides a partial answer.

Richardson, Vu, and Wu [11] prove that for d-dimensional convex bodies with C^2 boundary, the variance of the volume of the inscribed random polytope satisfies $\mathrm{Var} \asymp N^{-(d+3)/(d-1)}$, which gives N^{-3} for $d = 3$. Their methods involve Poissonization and yield unspecified constants; the explicit value 576π does not appear to follow from their framework.

4.3 Higher dimensions

The exact mean formula generalizes to all d: for N uniform points on S^{d-1}, the expected volume of the convex hull is

$$\mathbb{E}[\mathrm{vol}(P_N)] = \mathrm{vol}(B^d) \prod_{k=1}^{d} \frac{N-k}{N+k}.$$

The normalized leading term of the fractional deficit is $d(d+1)/N$, giving 12 for $d = 3$, 20 for $d = 4$, 30 for $d = 5$, and $d(d+1)$ in general. (Note: the pattern is $d(d+1)$, not $2 \cdot d!$ as might be guessed from the case $d = 3$ alone.)

However, the deficit scaling changes qualitatively for $d \geq 4$. For points on S^{d-1} with $d \geq 4$, the expected deficit scales as $N^{-2/(d-1)}$, not N^{-1}. (The exponent $2/(d-1)$ arises from the curvature of the sphere [6]; for $d = 3$ one has $2/(d-1) = 1$, recovering the N^{-1} scaling.) The product formula above

Table 2: Volume deficit statistics for $d = 4$. $N^{2/3}\overline{\Delta_N}$ estimates the mean constant; $N^{7/3}\widehat{\mathrm{Var}}$ tests the RVW variance scaling. The log-log slope of the variance converges to the predicted $7/3 \approx 2.333$.

N	$N^{2/3}\overline{\Delta_N}$	$N^{7/3}\widehat{\mathrm{Var}}$	Var. slope
100	39.96	167.5	—
200	42.67	236.9	1.83
500	44.89	305.5	2.06
1 000	45.89	340.2	2.18
2 000	46.52	363.2	2.24
5 000	47.02	380.2	2.28
10 000	47.24	390.3	2.30
20 000	47.38	391.7	2.33
50 000	47.48	399.4	2.31
100 000	47.53	408.3	2.30

gives exact rational expressions for all N, but the leading asymptotics for $d = 4$ become

$$\mathbb{E}[\Delta_N] \sim C_d \cdot N^{-2/3},$$

with convergence to the asymptotic regime that is much slower than in $d = 3$: the correction terms are $O(N^{-1/3})$ relative to the leading term, compared to $O(N^{-1})$ for $d = 3$.

We performed experiments for $d = 4$ with N up to 100,000. The rescaled deficit $N^{2/3}\Delta_N$ converges very slowly, reaching 47.53 at $N = 100{,}000$ (between the candidates $15\pi \approx 47.12$ and 48). The variance exponent, which Richardson, Vu, and Wu [11] predict to be $(d+3)/(d-1) = 7/3$ for $d = 4$, is confirmed by our data: the local log-log slopes of $\mathrm{Var}(\Delta_N)$ converge to 2.33 by $N = 20{,}000$. However, the variance *constant* has not yet converged at $N = 100{,}000$, reflecting the same slow-convergence phenomenon. Table 2 summarizes the $d = 4$ results.

4.4 Open questions

1. *Prove* Conjecture 1. The variance computation requires understanding the covariance structure of cap volumes across faces of the Delaunay triangulation, which goes beyond the techniques used for the mean in [1].

2. Compute the analogous variance constant for $d = 4$. Our experiments confirm the RVW exponent $7/3$ but do not yet pin down the constant. Is there a clean closed form analogous to 576π?

3. Characterize the rate of convergence in the CLT. Our data for $d = 3$ show persistent positive skewness $\sim N^{-1/2}$, suggesting that a Berry–Esseen bound of order $N^{-1/2}$ may be sharp.

4. Determine the full distribution of the normalized deficiency, answering the remainder of Open Question 5 of [1].

5 Ideal polytope volumes in hyperbolic 4-space

We now turn from Euclidean inscribed polytopes to ideal polytopes in hyperbolic space. An *ideal polytope* in $\mathbb{H}^4$ is the convex hull of N points on $S^3 = \partial\mathbb{H}^4$. Its hyperbolic volume can be computed by decomposing into ideal 4-simplices via Delaunay triangulation and summing their volumes using the Hopf–Poincaré–Schläfli angle formula.

5.1 Volume of an ideal 4-simplex

For a single ideal 4-simplex $[\infty, v_1, v_2, v_3, v_4]$ in the upper half-space model, the volume is an *exact affine function* of the ten codimension-2 dihedral angles:

$$\mathrm{Vol}_4(\Delta) = \frac{\pi}{3}\Big(4\pi - \sum_{k=1}^{10} \theta_k\Big). \tag{3}$$

The Euclidean tetrahedron $T = \mathrm{conv}(v_1, \dots, v_4)$ determines all ten angles: six are the Euclidean dihedral angles ϕ_{ij} of T, and four are $\arccos(d_\ell/R)$, where R is the circumradius and d_ℓ is the signed distance from the circumcenter to the face opposite v_ℓ.

Because formula (3) involves only arccos of algebraic numbers, the volume can be evaluated to *arbitrary precision* using extended arithmetic—no numerical integration is needed.

Remark 1. *The regular ideal 4-simplex has all ten dihedral angles equal to* $\arccos(1/3)$*, giving* $\mathrm{Vol}_4(\Delta_{\mathrm{reg}}) = \frac{\pi}{3}(4\pi - 10\arccos(1/3)) \approx 0.26890$.

5.2 Volume distribution for random ideal 4-polytopes

For N uniformly random points on S^3, we computed the total hyperbolic 4-volume of the corresponding ideal polytope (100,000 samples per vertex count, $N = 6$ to 100). Key findings:

1. The expected volume grows as $\mathbb{E}[V] \approx cN - b\sqrt{N}$, with $c \approx 1.10$ and a substantial $\sqrt{N}$ correction.

2. A beta distribution fits the marginal volume distribution better than a normal (KS statistics 0.002–0.005 for $N \geq 15$), mirroring the 3D behavior. However, the beta parameters α, β scale as $N^{1.40}$ and $N^{1.20}$ respectively, without the clean interpretation found in 3D.

3. The explanation lies in the *fluctuating Delaunay combinatorics*. In 3D, the number of Delaunay triangles for N points on S^2 is essentially deterministic ($2N - 4$ minus boundary). In 4D, the number of Delaunay tetrahedra for N points on S^3 has substantial variance: CV $\approx 13\%$ at $N = 10$, still 1.7% at $N = 100$.

4. The volume decomposes as $V = n \cdot \bar{v}$, where n is the number of Delaunay tetrahedra and $\bar{v}$ is the mean per-tetrahedron volume. Both $\bar{v}$ and n are approximately normally distributed, but they are *negatively correlated* ($\rho \approx -0.18$ at $N = 100$): denser triangulations produce more but thinner simplices.

5. Conditioned on n, the volume is well-described by a normal distribution (KS ≈ 0.02), while the beta fit fails. The per-tetrahedron volume $\bar{v} = V/n$ is also normally distributed. The marginal beta-like shape arises from the product of two negatively correlated normals, not from any intrinsic beta structure.

5.3 Optimal ideal 4-polytopes

For each N, we maximized the total hyperbolic volume over all configurations of N points on S^3, using L-BFGS-B with 200 random restarts and two-phase polishing. Table 3 summarizes the results.

The optimizer for $N = 8$ is the *16-cell* (4-cross-polytope), with f-vector $(8, 24, 32, 16)$ and $|\mathrm{Aut}| = 384 = 2^4 \cdot 4!$ (the hyperoctahedral group B_4). Using the exact coordinates $\pm e_i \in S^3$ and stereographic projection, each of the 8 congruent ideal 4-simplices has volume computable to arbitrary

Table 3: Optimal ideal 4-polytopes. The f-vector (f_0, f_1, f_2, f_3) counts vertices, edges, triangles, and tetrahedra of the boundary. |Aut| is the automorphism group order of the 1-skeleton (meaningful only when the skeleton is not the complete graph K_N, i.e., $N \geq 7$).

N	Volume	f-vector	|Aut|	Identification
5	0.26890	$(5, 10, 10, 5)$	120	Regular 4-simplex
6	0.58783	$(6, 15, 18, 9)$	720*	Neighborly (K_6)
7	1.02360	$(7, 19, 24, 12)$	48	
8	1.77143	$(8, 24, 32, 16)$	384	16-cell (hyperoctahedron)
9	2.31385	$(9, 29, 40, 20)$	80	
10	3.00910	$(10, 35, 50, 25)$	200	
12	4.37720	$(12, 46, 68, 34)$	8	
16	7.79862	$(16, 69, 106, 53)$	2	
20	11.9358	$(20, 92, 144, 72)$	16	
30	22.0037	$(30, 154, 248, 124)$	1	

*For $N \leq 6$, points on S^3 in general position form a neighborly polytope with 1-skeleton K_N, so $|\mathrm{Aut}(K_N)| = N!$ is not informative about geometric symmetry.

precision:

$$V_{\text{tet}} = \frac{\pi}{3}\Big(\frac{3\pi}{2} - 6\arccos\frac{1}{\sqrt{3}} + \arccos\frac{1}{3}\Big), \qquad V_{\text{16-cell}} = 8\,V_{\text{tet}} \approx 1.77143.$$

The extreme symmetry of the 16-cell also admits a direct computation in the Klein model: the hyperbolic volume element is $dV = d^4x\,(1-|x|^2)^{-5/2}$, and the 16-cell is the L^1 unit ball $\{x : \sum |x_i| \leq 1\}$, so the volume is 16 times a simplex integral. This yields the equivalent closed form

$$V_{\text{16-cell}} = \frac{4\pi^2}{3} - \frac{32\pi}{3}\,\arcsin\frac{1}{3} = \frac{4\pi^2}{15} - \frac{16}{5}\,v_4, \tag{4}$$

where $v_4 = \frac{10\pi}{3}\arcsin(1/3) - \frac{\pi^2}{3}$ is the volume of the regular ideal 4-simplex.

In contrast, the optimizer for $N = 9$ has $|\mathrm{Aut}| = 80$ with two vertex orbits (of sizes 4 and 5), degree sequence $[7^4, 6^5]$, and 10 congruent ideal 4-simplices involving exactly 5 distinct angle values (including $2\pi/5$ and $\pi/2$). However, all $\binom{9}{2} = 36$ pairwise inner products on S^3 are distinct: the graph automorphism group is purely combinatorial, not a metric symmetry. The two non-rational angles $\theta_1 \approx 51.03$ and $\theta_2 \approx 62.81$ satisfy $\cos(2\theta_1) = 2\cos^2\theta_1 - 1$ (accounting for the fifth angle type at 102.05), but do not appear

to be expressible in elementary closed form. No Klein-model reduction is available, and the volume $V(9) \approx 2.31385$ is accessible only through the angle formula or numerical optimization.

Comparison with symmetry-based constructions. An appealing strategy for constructing large-volume ideal 4-polytopes is to exploit the supplementarity $\arccos(1/3) + \arccos(-1/3) = \pi$ of the dihedral angles of the regular ideal 4-simplex and the ideal 16-cell: gluing a regular simplex onto one facet of the 16-cell produces a convex ideal polytope with 9 vertices and volume $V_{\text{16-cell}} + v_4 \approx 2.040$. While this exceeds the mean volume of a random 9-vertex configuration ($\mathbb{E}[V] \approx 1.08$), it falls 12% short of the actual optimum $V(9) \approx 2.314$. The optimizer achieves its advantage not through regularity of the individual simplices but through packing efficiency: 10 simplices at ≈ 0.231 each, versus 9 simplices (8 at ≈ 0.221 plus 1 at 0.269) in the glued construction. This illustrates a general principle: *maximal symmetry does not imply maximal volume*, and the gap between symmetry-based lower bounds and actual optima grows with N.

Surface area and isoperimetric ratio. Each boundary facet of an ideal 4-polytope is an ideal tetrahedron in a totally geodesic $\mathbb{H}^3 \subset \mathbb{H}^4$; its hyperbolic 3-volume can be computed by the classical Lobachevsky formula. The total 3-surface area $A = \sum_{\text{facets}} \text{Vol}_3(F)$ defines a natural companion to the 4-volume. Table 4 compares the maximizers of volume, surface area, and the isoperimetric ratio V/A for small N.

A striking finding: for $N \in \{5, 7, 8, 9, 10\}$, the same polytope simultaneously maximizes all three objectives (the entries marked "yes" in Table 4). These are precisely the highly symmetric optimizers—the regular simplex ($N = 5$), the 16-cell ($N = 8$), and the configurations with $|\text{Aut}| \geq 48$. At $N = 6$, the surface-area maximizer has 8 boundary facets all at maximum volume v_3 (the regular ideal tetrahedron in $\mathbb{H}^3$), while the volume maximizer has 9 facets at sub-maximal individual volume. For $N \geq 12$, all three objectives diverge, with the surface-area maximizer favoring fewer but more regular boundary tetrahedra.

Dihedral angles and rationality. For each optimizer, we examined whether the dihedral angles are rational multiples of π (which would be needed to tile $\mathbb{H}^4$ by copies of the polytope and obtain a hyperbolic manifold). The cosines of the dihedral angles are *algebraic numbers of low degree*—square roots of rationals—but the angles themselves are generically *not* rational

Table 4: Three optimization objectives for ideal 4-polytopes. For $N \leq 10$ (excluding $N = 6$), all three objectives are maximized by the *same* polytope. At $N = 6$ and $N \geq 12$ the optimizers diverge.

	Max Volume			Max Area			Max V/A	
N	V	A	V/A	A	V	V/A	V/A	same?
5	0.269	5.074	0.053	5.075	0.269	0.053	0.053	yes
6	0.588	7.818	0.075	8.119	0.538	0.066	0.075	V=iso≠A
7	1.024	11.83	0.087	11.83	1.024	0.087	0.087	yes
8	1.771	16.24	0.109	16.24	1.771	0.109	0.109	yes
9	2.314	19.96	0.116	19.96	2.314	0.116	0.116	yes
10	3.009	23.91	0.126	23.91	3.009	0.126	0.126	yes
12	4.377	32.33	0.135	32.43	4.015	0.124	0.136	all differ
16	7.799	50.49	0.154	50.74	7.785	0.153	0.155	all differ
20	11.86	70.35	0.169	70.58	11.81	0.167	0.168	all differ

multiples of π. In detail:

- $N = 5$: all angles $\arccos(1/3)$;
- $N = 6$: angles $\arccos(\sqrt{3/7})$, $\arccos(-1/7)$, $2\pi/3$;
- $N = 8$: angles $\arccos(1/\sqrt{3})$, $\pi/2$, $\arccos(-1/3)$;
- $N = 9$: includes $2\pi/5$ and $\pi/2$ alongside irrational-over-π angles.

Only $\pi/2$, $2\pi/3$, and $2\pi/5$ appear as rational multiples of π; the dominant angles are irrational over π in every case. This means these optimal ideal 4-polytopes cannot directly tile $\mathbb{H}^4$ to produce manifolds.

A sharp symmetry phase transition occurs around $N = 12$: for $N \leq 10$, the optimizers have large automorphism groups (48–384, excluding the trivial K_N cases), few distinct angle values (1–8), and all tetrahedra are nearly congruent. For $N \geq 12$, symmetry collapses to $|\mathrm{Aut}| \leq 16$, with hundreds of distinct angles and irregular structure.

5.4 Facet-parameterized optimization

The vertex-based optimization of Section 3 parameterizes ideal 4-polytopes by their vertices on S^3 and fixes the vertex count N. A complementary approach parameterizes by *facet normals*: given f unit vectors $u_1, \ldots, u_f \in$

Table 5: Facet-parameterized optimal ideal 4-polytopes. f denotes the number of facet normals, N the emergent vertex count, and V the hyperbolic 4-volume. The scan column gives the best volume from 100 random normal configurations (inner solve only); the optimized column includes the outer Powell optimization over normals.

f	N	V_{scan}	V_{opt}	Improvement	V/f^2
8	18	5.56	8.01	44%	0.125
10	30	12.37	15.92	29%	0.159
12	41	18.68	26.65	43%	0.185
16	61	36.71	41.51	13%	0.162
20	91	57.07	68.60	20%	0.172
24	111	78.42	84.27	8%	0.146

S^3, define

$$P(h) = \{x \in \mathbb{R}^4 : u_i \cdot x \leq h_i \text{ for all } i\}.$$

This polytope has at most f facets, with the number of vertices determined by the arrangement of the bounding hyperplanes. To obtain an *ideal* polytope, we require all vertices of $P(h)$ to lie on S^3, i.e., $|v|^2 = 1$ for each vertex v.

For a fixed combinatorial type (determined by the normals u_i at a probe value of h), each vertex is a linear function $v_j = A_j h$ of the support numbers, where A_j encodes the four active hyperplane constraints at vertex j. The ideal condition $|v_j|^2 = 1$ becomes a system of quadratic equations in the f unknowns h_i, which we solve by Levenberg–Marquardt to machine precision. This inner solver succeeds for 30–70% of random normal configurations, with the success rate depending on f.

Two-level optimization. To maximize volume over the facet normals, we use a two-level scheme: the outer loop optimizes the $3f$ hyperspherical angles parameterizing the normals on S^3 via Powell's method, while the inner loop solves the ideal constraint for each candidate set of normals. Table 5 summarizes the results.

Several features stand out. First, the outer optimization consistently improves volumes by 8–44% over the best random scan, demonstrating that the normal directions matter significantly. Second, the vertex count N is not fixed by f but emerges from the dual construction: the number of vertices equals the number of facets of $Q = \text{conv}(u_i/h_i)$, which varies with the normal

configuration. For simplicial 4-polytopes, N satisfies the Dehn–Sommerville relation $f_3 = f_1 - f_0$, and the typical vertex count grows roughly as $N \approx 4f$.

Third, the optimal volume scales approximately as $V \approx 0.17\, f^2$. The ratio V/f^2 in the last column is nearly constant across the range $f = 8$ to 24, suggesting a quadratic growth law. This should be contrasted with the vertex-parameterized scaling, which is closer to linear in N (cf. Table 3).

The cross-validated volumes agree to machine precision with those obtained from the independent general Schläfli formula for hyperbolic 4-simplices [12], which handles both compact and ideal vertices via Murakami's dilogarithm formula for spherical link volumes.

Acknowledgments

Computations were performed using `scipy.spatial.ConvexHull` with parallelization over 64 CPU cores. Code is available at `https://github.com/igorrivin/ideal_poly_volume_toolkit`.

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